

Nanocellulose mediated injectable bio-nanocomposite hydrogel scaffold-microstructure and rheological properties

Kapender Phogat · Sanchita Bandyopadhyay-Ghosh 

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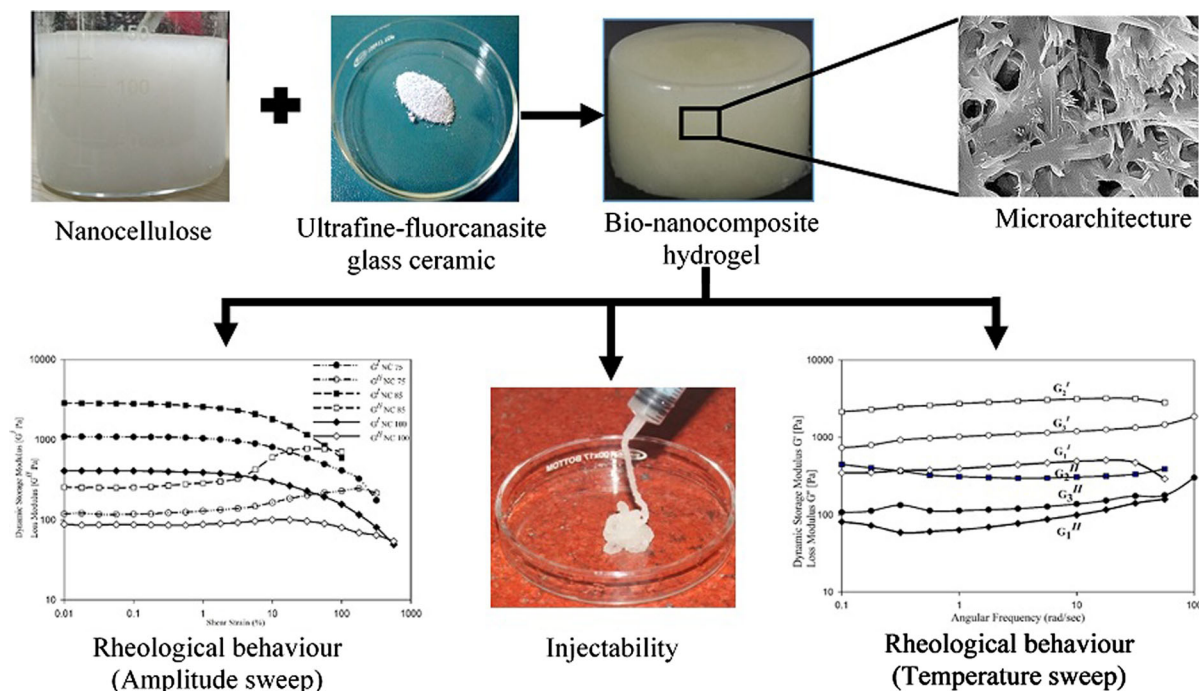
Abstract Here we report, for the first time, development of nanocellulose based novel bio-nanocomposite injectable hydrogel scaffold. An integrated approach has been used to incorporate ultrafine fluoranite (FC) glass-ceramic particulates within nanocellulose matrix, synthesized from lignocellulosic biomass. Epichlorohydrin (ECH) was used as a chemical cross-linking agent for hydrogel synthesis. The compositions and properties of bio-nanocomposite hydrogels were controlled by varying the loading of FC and ECH. Moreover, microstructural characterizations and functional group analysis of synthesized nanocellulose and bio-nanocomposite hydrogels were carried out using scanning electron microscopy (SEM) and Fourier transform infrared spectroscopy (FT-IR). SEM studies revealed successful extraction of cellulosic nanofibres and interconnected porous micro-architecture of injectable hydrogels. Incorporation of FC particulates resulted into development of well-

defined micro-porous morphology. FT-IR studies indicated the presence of hydroxyl groups and possible interactions between nanocellulose and FC reinforcements. The rheological study of the developed injectable hydrogel was carried out at different frequency, amplitude and temperature sweeps. The studies revealed that after the incorporation of FC, dynamic storage modulus increased, while synergistically enhancing the injectability of the bio-nanocomposite hydrogels. An optimum loading level of FC reinforcement could also be established. Swelling behaviour studies of the hydrogels demonstrated higher degree of swelling and enhanced structural integrity with incorporation of FC. The novel bio-nanocomposite injectable hydrogels developed in this study, could demonstrate promising potential as injectable implant material.

K. Phogat · S. Bandyopadhyay-Ghosh (✉)
Department of Mechanical Engineering, Manipal
University Jaipur, Jaipur, Rajasthan, India
e-mail: sanchitab.ghosh@jaipur.manipal.edu

K. Phogat
Department of Mechanical Engineering, JECRC
University, Jaipur, Rajasthan, India

Graphical abstract



Keywords Bio-nanocomposite hydrogel · Injectable scaffold · Nanocellulose · Fluorcanasite · Biomaterial

Introduction

The various types of scaffold materials used till date for hard tissue (such as bone) regeneration and repair are metallic, polymeric, ceramic and composite scaffold materials (Bandyopadhyay-Ghosh et al. 2008). Although, the metallic materials for scaffolds have excellent mechanical properties, they usually cause stress-shielding, owing to their non-similar modulus of elasticity to that of hard tissue (Alvarez and Nakajima 2009). Besides, these are prone to corrosion, have poor biological recognition on the material surface and are often known to release toxic ions. Polymeric scaffold materials, on the other hand, are easy to shape, lightweight, with the possibility to achieve tailored properties; they are not bioactive. Bioceramic scaffold materials on the other hand, mimic the inorganic bone composition, are bioactive and biocompatible (Bandyopadhyay-Ghosh 2008;

Bandyopadhyay-Ghosh et al. 2008; Gerhardt and Boccaccini 2010). However, due to the difficulty in casting complex shapes and their poor fracture toughness, use of these materials as scaffold materials alone is restricted. Composite materials, in this regard, can play a vital role. Development of an interconnected polymer-inorganic (ceramics, glass, glass-ceramic) composite scaffold takes advantages of both polymer and inorganic components to meet mechanical and physiological requirements of the host tissue (Fernandez-Yague et al. 2015). The newer composite scaffolds contain bioactive components, which are released from the implant providing a supply of biologically active moieties, that can be incorporated into the damaged tissue, thereby accelerating the healing process. From a biological perspective, it also makes sense to combine biopolymers and inorganics to fabricate composite scaffolds for bone tissue engineering because hard tissues are often the combination of a naturally occurring polymer and biological apatite (Bendtsen and Wei 2015).

An injectable composite system takes advantage of the above benefits of a composite system along with that of an injectable system, where a more homogeneous distribution of bioactive molecules within the

matrices can readily be obtained by virtue of the scaffold components being in suspension or solution before solidification *in vivo*. Injectable composite scaffolds is particularly interesting because the use of these materials avoid the surgical interventions to implant the scaffolds at defective/fractured areas. Once inside the body, the hydrogel scaffold can reassemble within a short interval and start recovery in bone formation (Jin et al. 2009; Naahidi et al. 2017). One of the promising injectable materials that can be used as scaffold is hydrogel (Temenoff and Mikos 2000; Nourmohammadi et al. 2016). Hydrogels are soft materials with a wide range of characteristics and applications areas. They have cross-linked hydrophilic networks that can absorb large quantities of water (or biological fluids) while maintaining their original structure (Ahmed and Aggor 2010; Jung et al. 2017; Portnov et al. 2017). When hydrogels are injected into an irregularly shaped defect, they can readily wet all surfaces of the injured site and create a low-density aqueous cavity that contains all the components necessary for tissue regeneration. Among all type of raw materials available for preparation of hydrogel, cellulose is the most abundant one, available worldwide and it combines hydrophilicity with good mechanical properties. Interestingly, the reinforcing potential of cellulose in the nanoscale dimension combines the benefits of bio-based materials and nanotechnology in a synergistic manner (Bandyopadhyay-Ghosh et al. 2014; Safdari et al. 2017). By extracting cellulose at the nanoscale, the majority of the defects associated with the hierarchical structure can be removed, and a new cellulose based “building block” becomes available. Cellulose nanoreinforcements also have high aspect ratio, low density, very low coefficient of thermal expansion and a reactive surface of –OH side groups that facilitates grafting chemical species to achieve different surface properties (surface functionalization) (Novak 1993; Nogi et al. 2005; Yano et al. 2005). However, there have been no reported study so far on use of nanocellulose as injectable hydrogel matrix. We anticipate that by combining injectable nanocellulose hydrogel matrix with a suitable inorganic filler would open a promising field for hard tissue replacement and repair using environmentally benign nanomaterial. These injectable bio-nanocomposite hydrogels, would only require minimally invasive interventions. Against this background, this paper reports for the first time,

development and characterisation of nanocellulose based injectable hydrogel system. We also report, for the first time, incorporation of fluorcanasite, a novel bioactive glass-ceramic reinforcement, in the nanocellulose hydrogel towards development of injectable bio-nanocomposite hydrogel. Microstructural and rheological properties of the bio-nanocomposite hydrogels were also investigated to primarily understand the potential of the novel hydrogels for scaffold applications.

Experimental

Materials

Wheat straw was purchased from the local suppliers. All chemicals like hydrogen peroxide (30%) (H_2O_2), epichlorohydrin 98.5% (ECH), sodium hydroxide (NaOH) pellets and urea (NH_2CONH_2) used in this study were procured from Sigma-Aldrich (India). Formic acid (HCOOH , 90%) was procured from Rankem-India. Silicon dioxide (SiO_2), calcium fluoride (CaF_2), sodium carbonate (Na_2CO_3), potassium carbonate (K_2CO_3), calcium oxide (CaO), calcium hydrogen phosphate (CaHPO_4), Hank's balanced salt solution and deionized water were procured from General Scientific, Jaipur, India.

Methods

Synthesis of nanocellulose

Wheat straw was used as raw material for nanocellulose synthesis. Chemo-mechanical method followed by ultrasonication was used for fibrillation and isolation of nanocellulose.

Synthesis of ultrafine fluorcanasite glass-ceramic

Fluorcanasite glass-ceramic has an excellent potential to be used as a bioactive implant as reported in literature by Bandyopadhyay-Ghosh et al. (2008). In the present study, ultrafine-fluorcanasite glass-ceramic powder was prepared by ‘Melt- Drive’ method. In this method, modified fluorcanasite glass batch was prepared using various oxides and fluorides. It was then melted in raising hearth furnace at a temperature of 1450 °C for 3 h (Bandyopadhyay-Ghosh 2008;

Bandyopadhyay-Ghosh et al. 2008, 2010). The molten glass batch was then poured into deionized water to form glass frits. The obtained glass frits were ball-milled to form glass powder. The glass powder was passed through a controlled two stage heat treatment followed by subsequent ball-milling to obtain ultrafine fluorcanasite glass-ceramic (Bandyopadhyay-Ghosh 2008; Bandyopadhyay-Ghosh et al. 2007, 2008, 2010). The synthesized ultrafine fluorcanasite glass-ceramic particulates were used as bioactive reinforcements for injectable nanocellulose based hydrogels.

Synthesis of injectable bio-nanocomposite hydrogel

Nanocellulose was used as matrix material whereas ultrafine fluorcanasite glass-ceramic powder was used as reinforcing material for the synthesis of injectable bio-nanocomposite hydrogel. Epichlorohydrin (ECH) was used as chemical cross linker. The synthesized nanocellulose was first dried in hot air oven at 60 °C and was then mixed with aqueous suspension of sodium hydroxide (NaOH, 90%) and urea followed by lyophilization at −20 °C for 12 h. Ultrafine fluorcanasite glass-ceramic particulates were then added to the above suspension at different loading levels. Water bath ultrasonication followed by the probe ultrasonication for 1 h was carried out for efficient dispersion of the reinforcement. Chemical crosslinker epichlorohydrin (ECH) was subsequently added to obtain the bio-nanocomposite hydrogel. These bio-nanocomposite hydrogels were then weighed and stored in refrigerator in the gel form.

Characterizations

Scanning electron microscopy (SEM)

Nova-Nano FE-SEM450 was used for microstructural characterization of the synthesized nanocellulose, fluorcanasite glass-ceramic and bio-nanocomposite hydrogels. Nanocellulose was sputtered with 5–10 nm thick gold coating before imaging using Quoron Q150R S sputter coater. Hydrogels were first frozen in lyophilizer at −20 °C for 48 h and then critically dried in a vacuum oven at 50 °C till its mass became constant. A scalpel was used to cut small sections of the hydrogels, which were subsequently sputtered with a 10–20 nm thick gold coating before imaging.

Fourier transform infrared spectroscopy (FT-IR)

Bio-nanocomposite hydrogels at different loadings of ultrafine fluorcanasite glass-ceramic particulates were analyzed with an FT-IR spectrophotometer (Bruker ALPHA-T) in the region of 5000–400 cm^{−1} in an attenuated total reflectance (ATR) mode to identify the functional groups present into the bio-nanocomposite hydrogels.

Viscoelastic properties

The rheological properties of hydrogels were studied using Anton Paar Rheometer MCR-702 with parallel plate geometry of 25 mm at room temperature. The gap distance between the plates was 1 mm. The disc shaped samples were adhered to the plates, and the initial force was kept below 1.0 N. Samples were compressed for 100% shear strain with an initial force of 0.59 N and a waiting time of 180 s to achieve the thermal gradient of the sample. A dynamic frequency sweep test from 0.1 to 100 rad/s, and temperature sweep test from 20 to 70 °C were performed to determine the dynamic storage modulus (G') and loss modulus (G'') of each hydrogel, at a strain rate confirmed to be in the linear viscoelastic range for each type of hydrogel prepared.

Injectability and swelling behaviour

The injectability test was carried out by applying manual shear using a syringe. About 1–2 ml of the control gel and other gels was loaded into a 5 ml syringe and a slow and controlled shear was applied. The flow was observed visually. The swelling degree (SD) of bio-nanocomposite hydrogels were measured at room temperature by immersing the hydrogel cubes in 30 ml of Hank's solution for a period of 0–40 days. First, the original dry weight of each sample was measured and noted. Small cubic sample of prepared bio-nanocomposite hydrogel (~ 0.2 grams) at dried state was then placed in the beaker containing the Hank's solution and weight gains were measured as a function of immersion time in 30 ml solution. The weight of submerged cubic piece of bio-nanocomposite hydrogel was noted on regular intervals after removing the surface solution with filter paper. The SD was calculated as below:

$$SD(\%) = \frac{W_s - W_d}{W_d} \times 100\%$$

where W_s and W_d are the weight of swollen hydrogel and dry hydrogel, respectively. The measurements were made at predetermined intervals of 0, 3, 7, 14, 30, 40 days.

Results and discussion

Scanning electron microscopy (SEM)

The SEM image as shown in Fig. 1a reveals successful extraction of nanoscale cellulosic fibres. The treatment of the fibres resulted in defibrillation of nanofibres from the cell walls and the SEM image of the cellulose fibres shows the separation of the nanofibres from the micro-sized fibres.

The diameter ranges from 20 to 50 nm with an average diameter of 35 nm of the synthesized nanocellulose. The SEM image of fluorcanasite glass-ceramic particulates shown in Fig. 1b reveals that the dimensions of particulates were ranging from 200 to 800 nm and were referred to as ultrafine.

Figure 2 shows the scanning electron micrographs of synthesized hydrogels. As shown in Fig. 2a₁–a₂, 100% nanocellulose based hydrogel demonstrated highly fibrous needle like structure, while Fig. 2b₁–c₂ show that with incorporation of increasing fluorcanasite content, the nanocomposite hydrogels

developed less fibrous (needle like) structure. A progressive increase in the structural density was visible. It is anticipated that while, fibrous structure would help in anchorage of cells, flattened and less fibrous morphology might prove to be beneficial for cell growth. Figure 2b₁–b₂ show that using a composition of 85% nanocellulose as matrix and 15% ultrafine fluorcanasite glass-ceramic particulates, a mixed mode of fibrous and non-fibrous morphologies could be obtained. However the structural morphology of the entire constructs was not uniform. Figure 2c₁–c₂ show the micrographs of nanocomposite hydrogel with 75% nanocellulose as matrix and 25% ultrafine fluorcanasite. As shown in the figure, the nanocomposite showed the least fibrous structure of all. From the micrographs, it is also clear that the developed hydrogels have interconnected pores which are anticipated to facilitate the supply of the nutrients and removal of waste materials during the bone healing or regeneration. A similar trend could be observed while comparing the magnified SEM images as shown in Fig. 2a₂–c₂ of the hydrogels. While, porous microarchitecture could be observed in all hydrogels, Fig. 2b₂ showed well defined interconnected pore structure along with three dimensional fibrous matrix, which would be beneficial for cell adhesion and growth in a scaffold.

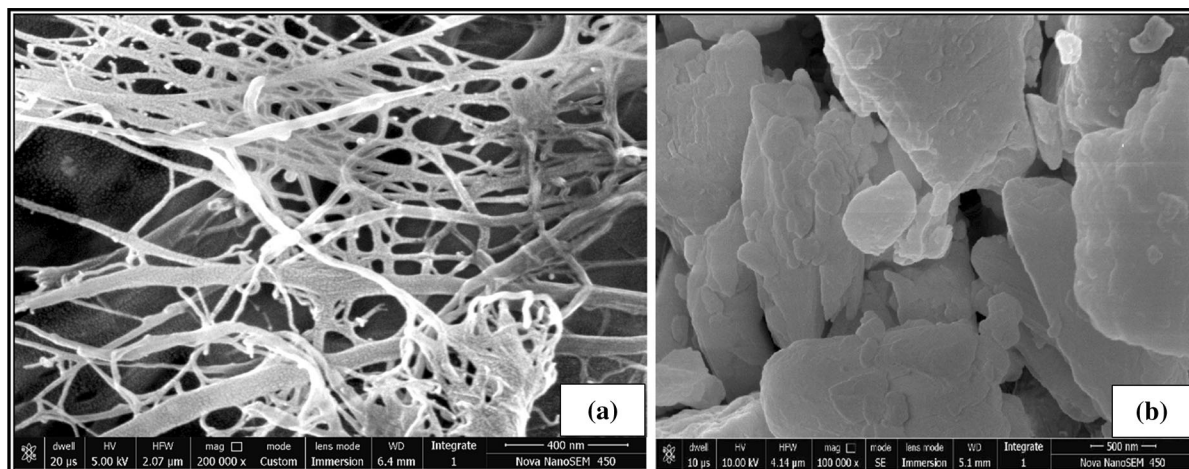


Fig. 1 Scanning electron micrograph of synthesized **a** nanocellulose (200000×) and **b** ultrafine fluorcanasite glass-ceramic particulates (100000×)

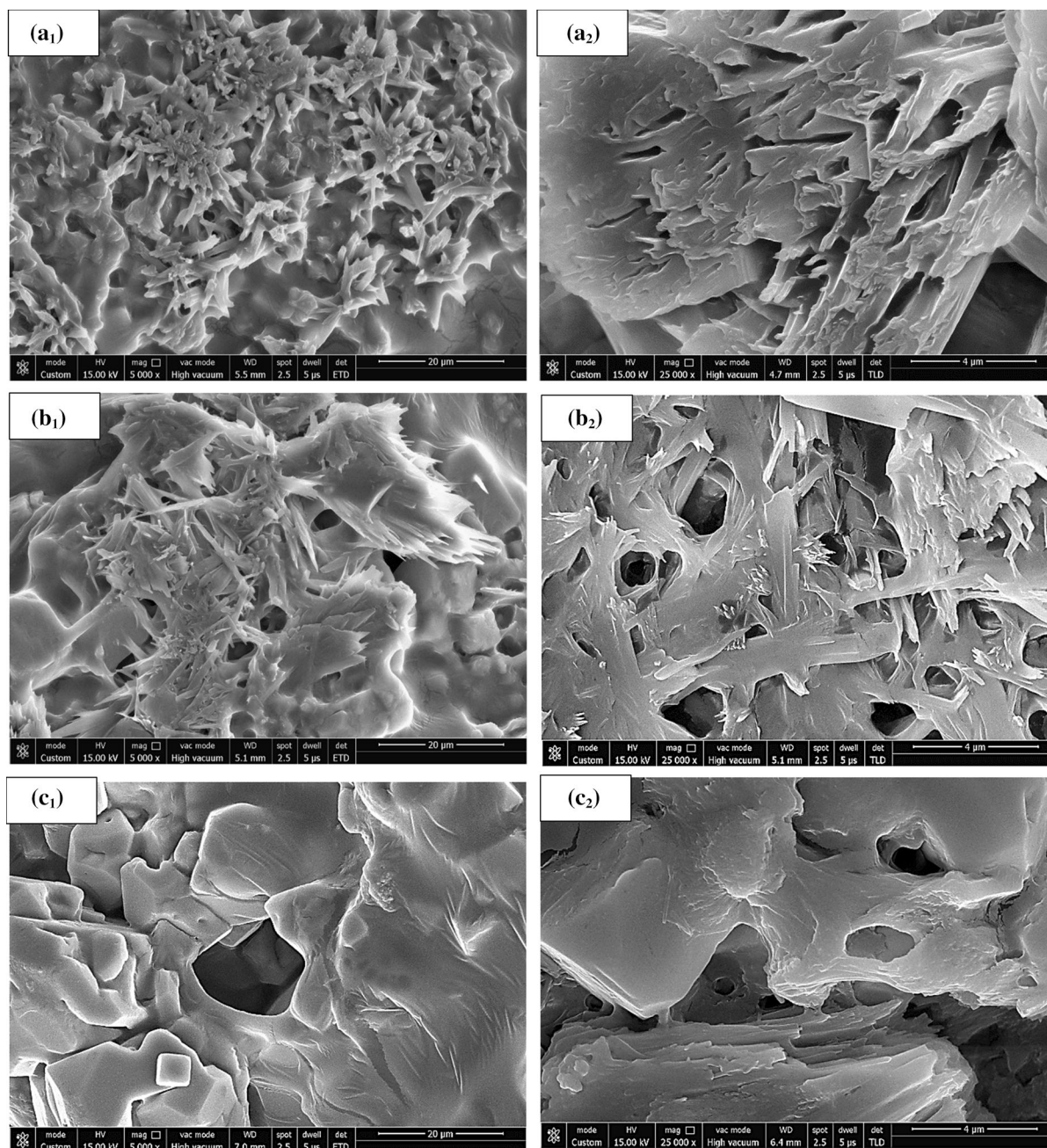


Fig. 2 Scanning electron micrographs of hydrogels. **a₁** Pure nanocellulose based hydrogel at $\times 5000$ magnification; **a₂** pure nanocellulose based hydrogel at $\times 25,000$ magnification; **b₁** nanocomposite hydrogel containing 85% nanocellulose and 15% ultrafine fluorcanasite glass-ceramic particulates at $\times 5000$ magnification; **b₂** nanocomposite hydrogel containing 85% nanocellulose and 15% ultrafine fluorcanasite glass-

ceramic particulates at $\times 25,000$ magnification; **c₁** nanocomposite hydrogel containing 75% nanocellulose and 25% ultrafine fluorcanasite glass-ceramic particulates at $\times 5000$ magnification and **c₂** nanocomposite hydrogel containing 75% nanocellulose and 25% ultrafine fluorcanasite glass-ceramic particulates at $\times 25,000$ magnification

FT-IR spectral analysis

The FT-IR spectra of nanocellulose based hydrogel and the hydrogels with different loadings of ultrafine fluorcanasite (FC) glass-ceramic are shown in Fig. 3a. In all cases, a broad and intense hump in range $3200\text{--}3600\text{ cm}^{-1}$ could be observed, which were attributed to hydrogen bonded hydroxyl group ($-\text{OH}$ group). The intensity of the hump increased as we increased the percentage loading of FC in bio-nanocomposite hydrogels. Another hump was present in the range of $1500\text{--}1800\text{ cm}^{-1}$ which can be attributed to the presence of $-\text{OH}$ bond from absorbed water owing to the hydrophilicity of nanocellulose. Another additional band in the range of $900\text{--}1200\text{ cm}^{-1}$ can be ascribed to C-O-C and C-O vibrations.

Viscoelastic properties

The viscoelastic properties of the developed bio-nanocomposite hydrogels with different loading of FC

were measured by applying the amplitude sweep Fig. 3b, frequency sweep Fig. 3c and temperature sweep Fig. 3d techniques described above. The yield stress of the tested samples shown in Fig. 3b–d varies from 140 to 170 Pa. In this analysis G' represents the dynamics storage modulus, which in turn represents the elastic part on behalf of the energy stored in the hydrogel. Correspondingly, G'' is the loss modulus that represents the viscous part on behalf of the energy dissipated in the hydrogel. Figure 3b–d clearly depict that the dynamic storage moduli of bio-nanocomposite hydrogels were significantly higher than that of nanocellulose hydrogel. Further, the results indicate that when the cellulose was used at nanoscale, the stability of the hydrogels improved significantly as compared to cellulose based hydrogel reported in the literature by Boonmahitthisud et al. (2016). The authors reported the presence of flow point (the point where dynamic storage modulus and loss modulus intersect) between 6 and 8% of shear strain in case of cellulose hydrogel; whereas our study has shown that flow point shifted to nearly 560% of the shear strain,

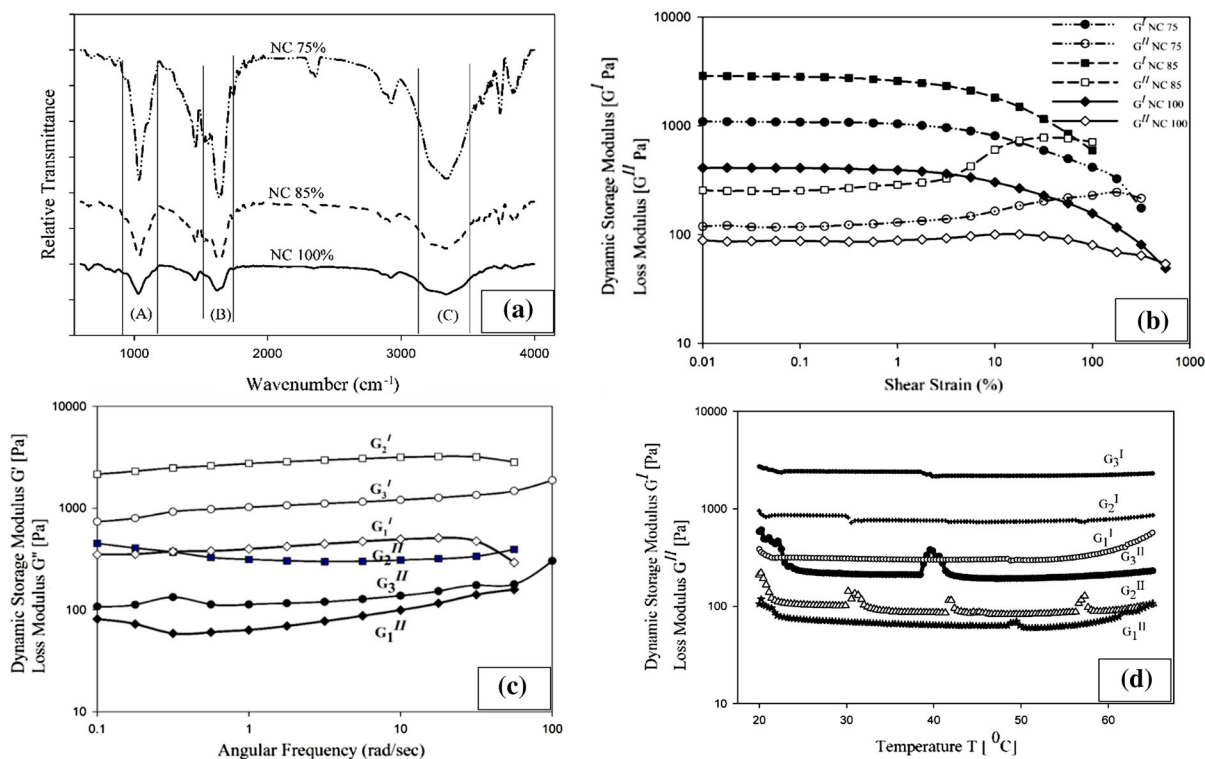


Fig. 3 a FT-IR spectra of hydrogels. Viscoelastic properties of hydrogels with b amplitude sweep, c frequency sweep and d temperature sweep

when cellulose was used at nanoscale, which in-turn would signify higher mechanical stability of hydrogels developed in this study. Meanwhile, for all the hydrogel samples, G' was found to be higher than G'' , clearly indicating the elastic behaviour of the hydrogels.

Figure 3b also reveals that with increased loading of ultrafine fluorcanasite glass-ceramic particulates, the base line for both the storage modulus and the loss modulus shifted above 1000 Pa, indicating higher stiffness, possibly owing to increased crosslinking. In this figure it is also noted that the flow point shifted towards left as and when we increase the loading of fluorcanasite glass-ceramic in the bio-nanocomposite hydrogel. This gives the clear indication of the flowability and in turn the injectability of the developed bio-nanocomposite hydrogels. The results also indicate that on incorporation of ultrafine fluorcanasite glass-ceramic particulates within nanocellulose matrix, lesser force would be required to get it injected and the injection pressure would be controlled by varying the loading level of reinforcements. Figure 3c depicts frequency sweep behaviour of different hydrogels. As shown in Fig. 3c, a frequency independent dynamic rheology behaviour was observed, as G' and G'' demonstrated little change in the experimental frequency range, which can be attributed to presence of strong covalent bonds in the hydrogels. Figure 3d reveals that the injectable hydrogels maintain their physical stability even at a temperature of 65 °C and hence can be stored easily before transplantation. Discontinuities were observed in few sections of loss modulus curves, which could be attributed to either due to variation in the gap between the parallel plates of the rheometer or owing to the presence of the moisture/air bubble.

Injectability and swelling behaviour

The injectability of the hydrogels was tested by loading them into a 5 ml syringe and subjecting them to manual shear. Figure 4 shows the good injectability of all the developed hydrogels. Figure 4 also indicates a shear-thinning (thixotropic) behaviour of all hydrogels, in which a gel is capable of forming a thick gel in situ when the shear force generated by injection is removed. Enhanced injectability was observed for the developed hydrogels while increasing the loading of FC particulates and same is supported by our rheological investigation using amplitude sweep as represented by Fig. 3b.

The results for the swelling behaviours of developed bio-nanocomposite injectable hydrogels in Hank's solution are shown in Fig. 5. In Fig. 5, plot (a) represents the variation in swelling behaviour of hydrogel containing 100% nanocellulose, whereas, plots (b) and (c) represent variation in swelling behaviours of 15 and 25% ultrafine fluorcanasite glass-ceramic particulate based injectable bio-nanocomposite hydrogels. As evidenced from Fig. 5, the degree of swelling increased with increased loading of fluorcanasite particulates. This could be attributed to increased porosity of the hydrogels upon incorporation of particulate phase, leading to higher degree of swelling. The results are also in good agreement with our FT-IR observations. It is important to note that the bio-nanocomposite hydrogels demonstrated higher resistance to disintegration, which could be attributed to enhanced structural integrity of the nanocomposite samples with incorporation of fluorcanasite particulate phase.

This is also in accordance with our results of viscoelastic analysis of bio-nanocomposite hydrogels.

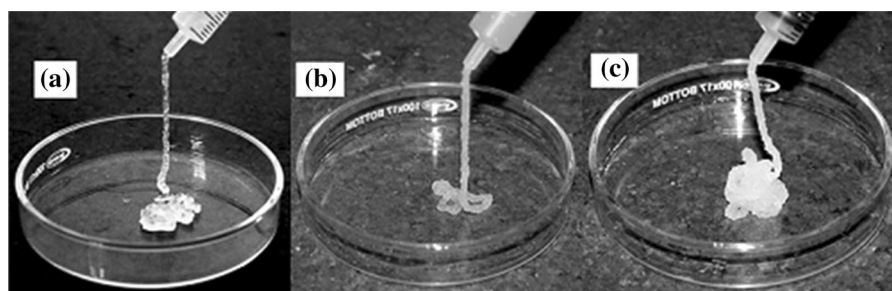


Fig. 4 Injectability test of hydrogels **a** pure nanocellulose based hydrogel, **b** nanocomposite hydrogel containing 85% nanocellulose and 15% ultrafine fluorcanasite glass-ceramic

particulates and **c** nanocomposite hydrogel containing 75% nanocellulose and 25% ultrafine fluorcanasite glass-ceramic particulates

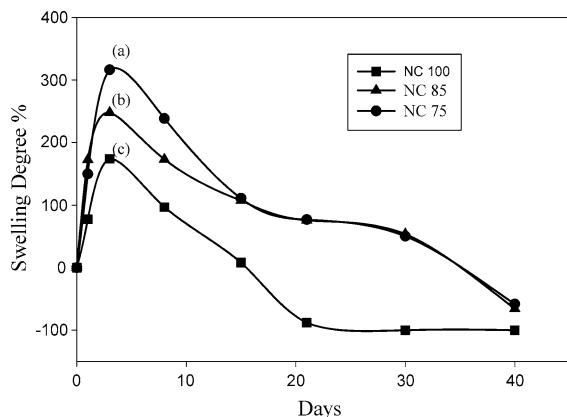


Fig. 5 Swelling behavior of hydrogels

It can be noted that while pure nanocellulose hydrogel collapsed in Hank's solution after 15 days, bio-nanocomposite hydrogels maintained their structural integrities even after 35 days. This could also mean that higher washout resistance is anticipated in vivo upon incorporation of fluorcanasite particulates, making them suitable as injectable scaffolds.

Conclusions

In our present study, we have successfully developed the novel injectable bio-nanocomposite hydrogels based on the agricultural biomass (wheat straw) derived nanocellulose and ultrafine fluorcanasite glass-ceramic particulates. The developed hydrogels showed self-standing property and injectability. SEM results indicated that the constituent cellulose was at nanoscale whereas the fluorcanasite glass-ceramic particulates were at ultrafine level. The hydrogels were found to have interconnected porous microarchitecture. 100% nanocellulose based hydrogel demonstrated highly fibrous needle like structure, while with incorporation of FC, a progressive increase in the structural density was apparent. From the micrographs, it was also clear that although, porous microarchitecture could be observed in all hydrogels, 15 wt% ultrafine fluorcanasite glass-ceramic particulates resulted into well defined interconnected pore microstructure along with three dimensional fibrous matrix, which is anticipated to help in cell adhesion and growth. FT-IR studies revealed the presence of the functional groups in the synthesized hydrogels and

confirmed the incorporation of FC in the bio-nanocomposite hydrogel. Rheological studies with different sweeps like amplitude, angular frequency and temperature sweep tests revealed that the dynamic storage modulus (G') and loss modulus (G'') increased with increase in the loading level of FC particulates which in turn, indicated increased dynamic stability of the developed hydrogels. It was noted that the value of G' increased with increasing in FC glass-ceramic particulate loadings, indicating higher stiffness in the nanocellulose matrix and effective increase in the elastic properties of the bio-nanocomposite injectable hydrogels, possibly due to the additional crosslinking points provided by FC glass-ceramic particulates. Moreover, the intersection point of G' and G'' represents the flow point or gel point of the samples that was shifted towards the low strain range with increase in FC glass-ceramic particulate contents, that further made the hydrogels injectable with minimal effort. An optimum loading level (15 wt%) of FC particulates could be identified, for obtaining improved mechanical properties and injectability, which was in good agreement with SEM observations. The bio-nanocomposite hydrogel therefore promises to be an excellent candidate material for injectable scaffold.

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